



University of Sri Jayewardenepura
Department of Computer Science

Project Report
on
IBM HR Analytics Employee Attrition & Performance

Submitted by:

Name : N.L.M.Arachchi
Index No : AS2021926
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Under the guidance of:

Dr. T.M.K.K. Jinasena
Senior Lecturer
Department of Computer Science

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DECLARATION

I, N.L.M.Arachchi hereby declare that the project work titled "IBM HR Analytics Employee Attrition & Performance" is a record of original work carried out by me under the guidance of Dr. T.M.K.K. Jinasena. This project has not been submitted to any other institution or university for the award of any degree, diploma or certification.

Navodya

14.11.2024

N.L.M.Arachchi

ACKNOWLEDGEMENT

I would like to express my sincere gratitude for the opportunity to work on this HR Attrition Data Project. This project has been a valuable learning experience, allowing me to deepen my understanding of data analysis and HR-related insights. I am thankful for the resources and tools that supported my research and analysis. This project was completed independently, and I take full responsibility for its content and execution.

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INTRODUCTION

Employee turnover is often perceived as detrimental to organizational efficiency, but is this always the case? This research explores the multifaceted factors influencing employee attrition and examines whether lower turnover invariably leads to greater efficiency. Utilizing the "IBM HR Analytics Employee Attrition and Performance" dataset, which includes variables such as employee age, department, education level, job satisfaction, gender, job role, marital status, and overtime hours, we conduct a comprehensive descriptive analysis to predict employee retention. By understanding these factors, HR and management can make informed decisions to mitigate attrition. This study aims to identify novel strategies to reduce employee turnover, providing actionable insights and predictions to help IBM retain talent and maintain productivity and success.

PROJECT OBJECTIVES

- Analyze historical employee data to identify key factors influencing attrition.

The dataset was cleaned and explored using visualizations and correlation analysis. Key factors such as OverTime, MonthlyIncome, and JobSatisfaction were identified as major contributors to attrition. Employees with low income, high overtime, and low satisfaction had higher chances of leaving

- Develop predictive models to classify employees likely to leave.

After preprocessing, three classification models—Logistic Regression, Random Forest, and Decision Tree—were trained. The Decision Tree model was selected for its simplicity and performance, and used to predict whether new employees are likely to leave.

DATASET DESCRIPTION

The dataset used for this project is the IBM HR Analytics Employee Attrition & Performance Dataset, available on Kaggle. It contains records of employees, including demographics, job roles, satisfaction levels, salary, and tenure details. The dataset has 35 features, such as

'Age'

'JobRole'

'OverTime'

'MonthlyIncome'

'BusinessTravel'

'Department'

'DistanceFromHome'

'Education'

'TotalWorkingYears'

'WorkLifeBalance'

'YearsAtCompany'

'YearsSinceLastPromotion'

'YearsInCurrentRole'

'TrainingTimesLastYear'

'Attrition', which indicate whether an employee has left the company.

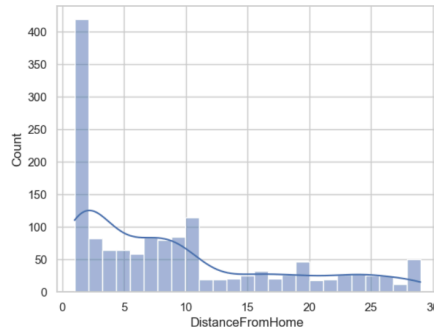
PROCESS WORKFLOW

◆ Step 1: Basic Information and Data Cleaning

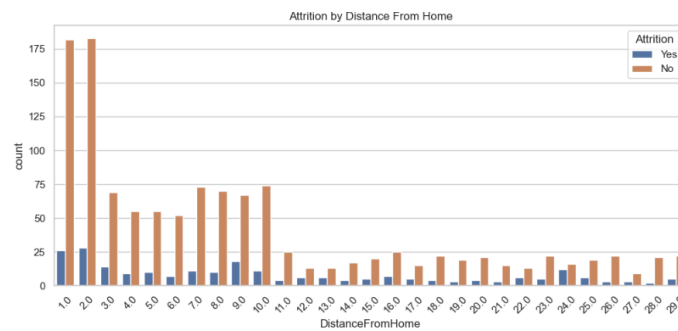
- Imported necessary libraries such as `pandas` for data handling.
- Loaded the dataset using `read_csv()` and viewed the first five records to understand the structure.
- Used `.info()`, `.describe()`, and `.shape` to get an overview of data types, summary statistics, and dataset dimensions.
- Checked for missing values and dropped rows or fields containing them.
- Identified and removed duplicate records to ensure data integrity.
- Dropped irrelevant columns such as `EmployeeCount`, `Over18`, `StandardHours`, and `EmployeeNumber`.
- Used boxplots to visualize numerical features and detect potential outliers.

◆ Step 2: Exploratory Data Analysis (EDA)

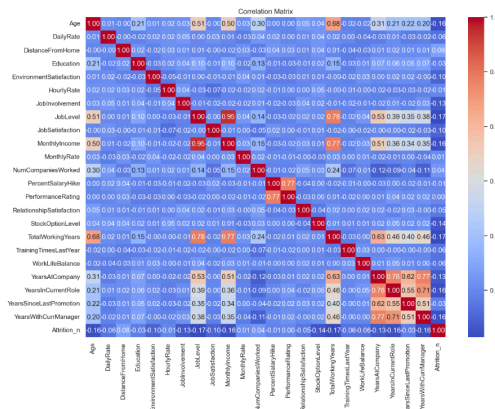
- Imported `matplotlib.pyplot` and `seaborn` for data visualization.
- Performed univariate analysis (e.g., distribution of age, income).



- Conducted bivariate analysis to explore relationships between features (e.g., Attrition vs. OverTime).



- Created heatmaps and pairplots to observe multivariate relationships



- Identified key features correlated with employee attrition.

◆ **Step 3: Data Preparation**

- Converted categorical variables (e.g., JobRole, Department, OverTime) into numerical format using encoding techniques.
- Ensured all data was in a suitable format for model training.

◆ **Step 4: Model Building**

- Split the dataset into features (X) and target (y).
- Trained three machine learning classification models:
 - Logistic Regression
 - Decision Tree
 - Random Forest
- Compared model outcomes and selected the Decision Tree model based on simplicity and performance.

◆ **Step 5: Prediction**

- Applied the final Decision Tree model to predict attrition on a new employee dataset.
- Provided a binary output indicating whether the employee is likely to leave or stay.

TOOLS AND TECHNOLOGIES USED

♦ Programming Language

Python – Used for data manipulation, visualization, and model building due to its powerful libraries and ease of use.

♦ Data Analysis & Processing Libraries

- Pandas – For data loading, cleaning, and manipulation.
- NumPy – For numerical operations and handling arrays.
- Scikit-Learn – Used for preprocessing, model training, and evaluation.
- Matplotlib & Seaborn – For creating visualizations such as boxplots, heatmaps, and bar charts.

♦ Machine Learning Algorithms

- Logistic Regression – Used for baseline classification.
- Decision Tree – Selected final model due to its interpretability.
- Random Forest – Used for comparison and analysis of prediction performance.

♦ Development Environment

- Jupyter Notebook – Used as the primary environment for coding, analysis, and visualization.

CONCLUSION

This project successfully analyzed the IBM HR Analytics – Employee Attrition & Performance dataset to identify key factors influencing employee turnover. Through data cleaning, exploratory analysis, and model development, important patterns were discovered—such as the impact of overtime, low income, and job dissatisfaction on attrition.

Machine learning models including Logistic Regression, Random Forest, and Decision Tree were trained to predict employee attrition, with the Decision Tree model selected for its balance of performance and interpretability. The insights gained can support HR teams in taking proactive steps to retain employees by addressing workload, compensation, and workplace satisfaction.

Overall, this project demonstrates how data-driven approaches can enhance decision-making in human resource management and contribute to reducing organizational turnover.

Thank You !

AS2021926

N.L.M.Arachchi